

# PIPE-Cypher: Automatic Enterprise Benchmark Generation for Text-to-Cypher Systems

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## Abstract

Enterprises increasingly use property graphs and Cypher to analyze fraud, risk, identity, customer, and operational data. A useful Text2Cypher benchmark for this setting must be private, refreshable, executable, and sensitive to graph-specific details such as relationship direction, schema vocabulary, and exact literal values. We present PIPE-Cypher, a local-model pipeline that generates NL-to-Cypher benchmarks inside an organization’s compute boundary. PIPE-Cypher profiles the graph, grounds question slots by executing reverse Cypher queries, generates constrained queries, checks them with deterministic Cypher validators, executes them, redacts value-bearing artifacts, and then applies a local LLM judge. With local Qwen3.5-9B generation and judging, PIPE-Cypher exports 3,000 accepted FinBench/SNB examples, completes three audited ablation suites, calibrates the automated judge, onboards ICIJ Offshore Leaks as a third public graph, and evaluates 11 completed local downstream models. The benchmark is hard in zero-shot settings; a strict no-signature few-shot control shows that schema-specific examples can help compatible model families, while same-signature example banks are reported only as an operational upper bound.

For evaluation, that privacy boundary matters as much as model accuracy. A static public dataset is useful for shared comparison, but it cannot contain a bank’s account taxonomy, an identity team’s permission graph, or the categorical values that make a query answerable. It also cannot change when the production schema changes. Industry teams therefore need something different: a repeatable way to turn a live property graph into a balanced, executable, privacy-aware NL-to-Cypher benchmark without sending sensitive schema or values to paid generation APIs.

PIPE-Cypher addresses this setting. The pipeline profiles a target graph, finds graph values that make candidate questions answerable, constrains a local generator, validates and repairs the generated Cypher, executes it, and then asks a local LLM judge to review only candidates that already have execution evidence. We treat Cypher correctness as something to check, not something to hope the prompt induces: relationship direction, read-only safety, exact literal use, categorical values, contextual return columns, and `RETURN DISTINCT` are enforced before examples are accepted.

## 1 Introduction

Property graphs are attractive in enterprise settings because the facts of interest are often relational paths: account transfers, identity entitlements, access chains, customer interactions, supplier dependencies, and fraud rings. Cypher gives analysts a compact language for these patterns. As LLMs become natural-language interfaces for graph analytics, an organization cannot evaluate a Text2Cypher system only on public schemas; it needs to know whether the model handles its labels, relationship directions, values, governance rules, and recurring operational questions.

**Contributions.** We make four contributions: (1) a local-model workflow for generating private NL-to-Cypher benchmarks from an organization’s own graph; (2) outcome-aware reverse grounding and Cypher-specific validators for read-only safety, relationship direction, exact literals, categorical values, contextual returns, and conservative rewrites; (3) a scaled public-proxy evaluation over FinBench, SNB, and ICIJ with ablations, judge calibration, redaction audits, and an 11-model local transfer study; and (4) reproducibility artifacts for onboarding, value sampling, benchmark refresh, evidence packaging, and appendix-level audit.

## 2 Related Work

Recent Text2Cypher resources, including Mind the Query (Chauhan et al., 2025), SyntheT2C (Zhong et al., 2025), Auto-Cypher (Tiwari et al., 2025), Text2Cypher (Ozsoy et al., 2025), Cypher-Bench (Feng et al., 2025), and the public Text2Cypher-2024 corpus (Neo4j, 2024), show that good Cypher data needs schema grounding, execution checks, verification, and complexity-aware evaluation. We do not claim that template filling or execution filtering is new by itself. The contribution is the way these pieces are made usable for enterprise benchmark generation. PIPE-Cypher adds outcome-aware reverse grounding before natural-language realization, deterministic Cypher governance before export, local judge calibration after execution, explicit privacy and value policies, and provenance-rich refresh artifacts for organizations that need to benchmark their own graphs.

Mind the Query is especially relevant because it reports an Industry Track Text2Cypher dataset with schema, runtime, value, and human logical validation. PIPE-Cypher keeps that validation discipline but changes the object of study. We are not proposing one more static dataset or tuning corpus. We are proposing the process an organization would use to generate, refresh, audit, and redact a benchmark for its own graph, local model endpoint, and value policy.

Text-to-SQL benchmarks such as Spider 2.0 (Lei et al., 2024) and BIRD (Li et al., 2023) have pushed text-to-query evaluation toward realistic database tasks and execution-based scoring, while execution-guided decoding shows why query execution is useful semantic feedback rather than a cosmetic check (Wang et al., 2018). Recent residual-skill Text-to-SQL work further shows that optimizing complementary agent skills on residual failures can improve selected accuracy across SQL dialects (Zhu et al., 2026). Graph query generation adds different failure modes: relationship direction can invert the meaning of a query, node and relationship properties live in different namespaces, and path-shaped questions can be syntactically valid while semantically ungrounded. CIKM AutoQuery (Zheng et al., 2024) motivates treating workload generation as a separate object of study from downstream model quality. LDBC FinBench (Qi et al., 2023) and SNB (Puroja et al., 2023) provide public graph workloads with financial and social-network structure, while

ICIJ Offshore Leaks gives an additional public finance/compliance onboarding check.

For benchmark quality, lexical diversity alone is not enough. A question bank can use varied wording and still overuse the same graph values or the same Cypher template. PIPE-Cypher therefore combines text-generation diagnostics such as Distinct-n (Li et al., 2016) and self-BLEU-style redundancy checks (Zhu et al., 2018) with text-to-query structure metrics: schema coverage, relationship/property coverage, query-signature diversity, structural feature rates, and normalized entropy over graph/category/difficulty cells. For subset construction, we use an MMR-style novelty objective (Carbonell and Goldstein, 1998) over Cypher signatures, template families, structural substructures, schema atoms, values, and question tokens.

## 3 Method

PIPE-Cypher has six stages: schema profiling, workload planning, reverse Cypher grounding, constrained generation and repair, deterministic validation and execution, and LLM-judge review. The central design choice is simple: accepted examples should prove that they are answerable and safe. Prompts can ask a model to respect relationship directions or exact literals, but accepted examples must pass schema checks, parser-style structure extraction, live execution, and judge review.

Schema profiling records labels, relationship types, properties, observed directions, and bounded low-cardinality categorical values. Workload planning targets eight categories that appear in operational graph analytics: simple retrieval, complex retrieval, simple aggregation, complex aggregation, boolean existence, negation/difference, path/temporal transaction, and ranking/top-k. Reverse grounding then runs read-only Cypher to find slot values that actually produce rows. This step avoids a common synthetic-data failure: a plausible-looking question that has no answer in the graph.

The deterministic layer checks read-only safety, syntax shape, labels and properties from the schema, explicit relationship types, observed relationship directions, schema-provided categorical values, and non-empty execution where required. Live execution uses read-only credentials and read-access sessions; token-level write rejection is only the first safety check. A lightweight Cypher analyzer extracts return aliases, variables, labels, re-

relationship patterns, risky constructs, and rewrite skip reasons. This makes normalization auditable instead of a silent string edit. The direction gate reads both outgoing and incoming arrow syntax before schema checking, and rejects undirected relationship patterns when direction is required.

## 4 Implementation

We implement PIPE-Cypher as a Python package. Neo4j is the experimental backend for this study, but the contribution is about Cypher and property graphs rather than a single vendor. Reported benchmark generation and judge review use a local Qwen3.5-9B endpoint behind a vLLM/OpenAI-compatible interface. Downstream evaluation separately tests 11 completed locally served checkpoints spanning general instruction, code-tuned, Cypher-tuned, and Text2Cypher-tuned families. Benchmark generation and evaluation run inside the organization’s compute boundary without paid generation APIs.

The built-in FinBench profile is grounded in the public datagen snapshot export and includes typed node and relationship properties. Live schema inspection also records low-cardinality string values as categorical constraints for prompting and validation. The generated Neo4j import script uses uniqueness constraints for nodes and creates, rather than merges, transaction relationships so repeated account-to-account events are preserved. The SNB reference profile is grounded in the official Neo4j/Cypher headers and read-query files.

For generation, PIPE-Cypher starts from workload templates whose slots are filled by reverse-binding queries. It can also use a mixed mode in which the LLM proposes additional templates after seeing proven workload seeds. Every run records both accepted and rejected candidates. Retrieved few-shot examples replace graph-specific values with typed placeholders, so the model sees the query structure without repeatedly seeing the same tenant values. A lightweight value grounder adds typed annotations for categorical values and reverse-bound entities, including punctuation variants, possessives, plurals, synonyms, name partials, and small typos.

Inspired by Mind the Query’s prompt-setting analysis, PIPE-Cypher exposes prompt profiles for schema-only, instructions-only, examples-only, examples-plus-instructions, and full governed generation. We report a profile only when it passes the

same target-size and evidence checks as the main run.

For LLM-judge review, we do not send the entire schema when a query touches only a small part of it. The judge prompt includes the labels, relationship types, and properties mentioned by the candidate query, while deterministic validators still check against the full schema. This keeps local 9B prompts manageable on larger graphs without weakening schema validation.

The Cypher layer uses constraints drawn from production Text2Cypher work: schema-only prompting, exact matching, relationship direction discipline, `RETURN DISTINCT`, reserved variable rejection, categorical values, required contextual return columns, fuzzy value annotations, placeholderized retrieval examples, and parser-aware rewrite boundaries. PIPE-Cypher records parser-style structure features and skips rewrites for risky constructs such as `UNION`, `CALL`, `UNWIND`, `WHERE EXISTS`, multiple `WHERE` clauses, or reserved variables.

We also estimate seed-template capacity before full generation. This check caught and fixed a scale blocker: reverse-binding execution was hard-capped to 10 rows, and no-slot negation/ranking seeds could not support full category targets. PIPE-Cypher now uses the configured binding limit and includes slotted negation and ranking seeds for both graphs.

For enterprise onboarding, PIPE-Cypher includes a deployment template for a company’s own graph: read-only credentials, local model endpoint, schema introspection, privacy policy, dry run, scaled run, audit, and redacted export. We validate this pattern on three public proxy graphs rather than a proprietary tenant deployment. Configurable value-sampling policies decide which low-cardinality graph values may enter prompts. Redacted exports replace quoted literals, entity values, and string-valued result samples with stable placeholders for broader internal review. The ICIJ onboarding run also pushed the implementation beyond hand-authored LDBC seeds: PIPE-Cypher now derives relationship-count, anti-join, and top-k templates from observed labels, relationship directions, and safe low-cardinality properties, then grounds slot values with outcome-aware reverse Cypher.

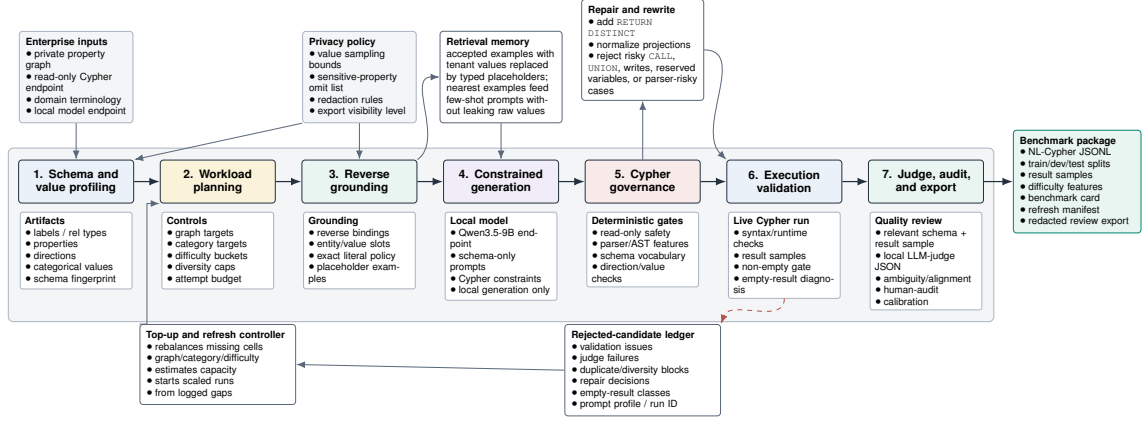


Figure 1: PIPE-Cypher benchmark generation pipeline. The key industry additions beyond static Text2Cypher dataset construction are privacy/value policies, reverse grounding, Cypher governance, execution diagnostics, local judge calibration, and benchmark export/refresh.

| Gate                              | Evidence checked                                           | Failure mode caught                       |
|-----------------------------------|------------------------------------------------------------|-------------------------------------------|
| Read-only safety                  | blocked Cypher write/admin tokens                          | destructive or operational queries        |
| Schema validity                   | labels, relationship types, properties, categorical values | hallucinated graph vocabulary or values   |
| Direction validity                | observed relationship directions                           | reversed or invalid traversals            |
| Cypher analysis and normalization | structure extraction, rewrite safety, RETURN DISTINCT      | unsafe rewrites or duplicate/noisy rows   |
| Question constraints              | quoted values use exact literals                           | fuzzy matching of exact user values       |
| Execution                         | query runs and returns rows                                | syntactic validity without answerability  |
| LLM judge                         | ambiguity, semantic alignment, schema use                  | executable but semantically weak examples |

Table 1: PIPE-Cypher candidate acceptance gates.

| Graph    | Target/category | Binding limit | Min seed capacity | All categories meet target |
|----------|-----------------|---------------|-------------------|----------------------------|
| FinBench | 250             | 300           | 300               | Yes                        |
| SNB      | 125             | 200           | 200               | Yes                        |

Table 2: Built-in seed-template capacity after removing the reverse-binding 10-row cap and adding slotted negation/ranking seeds. Capacity is a theoretical scale-readiness check; final examples still must pass validation, execution, diversity, and judge gates.

## 5 Experiments

**Research questions.** We evaluate four questions that matter for an industry benchmark generator: RQ1, can a local-model pipeline produce a balanced executable benchmark over live property graphs? RQ2, do Cypher-specific validation and grounding steps make generation reliable at scale? RQ3, does the resulting benchmark expose meaningful downstream Text2Cypher failures rather than merely checking syntax? RQ4, can the same workflow onboard a new public enterprise-style graph without hard-coding FinBench or SNB?

The generated benchmark contains 3,000 accepted examples: 2,000 from LDBC FinBench and 1,000 from LDBC SNB, balanced over eight work-

load categories. We report three completed FinBench/SNB ablation suites: target-50, corrected target-100, and a seed-17 target-50 repeat. Downstream Text2Cypher evaluation uses live execution accuracy and answer-set F1 as primary metrics; reference-based text metrics are supported for debugging only and reported in the appendix. We additionally report ICIJ Offshore Leaks as a third public finance/compliance onboarding proxy.

## 6 Results

**RQ1: executable benchmark generation.** The full live run produced 3,000 accepted examples from 4,925 candidates using local Qwen3.5-9B for generation and judging. Category-specific recovery top-ups filled the only under-target categories from the initial sequential run. Every exported example passed read-only, syntax, schema, execution, non-empty result, and judge gates.

The accepted records are exported with stable identifiers, train/dev/test splits, result samples, gate metadata, aggregate statistics, and a manifest hash; the appendix gives the full artifact distribution.

**Diversity and residual concentration.** We do

| Graph    | Candidates | Accepted | Acceptance | Categories at target |
|----------|------------|----------|------------|----------------------|
| FinBench | 3,405      | 2,000    | 0.587      | 8/8                  |
| SNB      | 1,520      | 1,000    | 0.658      | 8/8                  |
| Total    | 4,925      | 3,000    | 0.609      | 16/16                |

Table 3: Full live generation with local Qwen3.5-9B. Candidate counts include the initial sequential run and category-specific recovery top-ups.

not reduce diversity to one score. The full export has perfect category balance, near-perfect difficulty balance, 1,115 unique grounded entity values, and exact quotation of grounded values in 82.6% of examples with entity bindings. Query-signature diversity remains low because seeded graph-grounded templates carry much of the generation load. Table 4 shows that diversity is still governable after acceptance: at the same graph/category target, a selector using query signatures, template families, structural substructures, schema atoms, values, and question tokens improves structural coverage, adjusted Distinct-2, query-signature ratio, and property coverage. The result is a balanced executable benchmark, plus clear diagnostics for what should diversify during refresh.

| Metric                       | Random | Governed |
|------------------------------|--------|----------|
| PIPE-Diversity index         | 0.557  | 0.575    |
| Unique query-signature ratio | 0.062  | 0.135    |
| Structural substructures     | 97     | 134      |
| Adjusted Distinct-2          | 0.266  | 0.284    |
| Property coverage            | 0.407  | 0.426    |

Table 4: Diversity-governed selection at the same graph/category target. The selector improves structural, lexical, signature, and schema coverage after all quality gates have already passed; the appendix reports the full diversity audit, including residual template concentration.

**RQ2: governed generation.** Most full-run rejections come from duplicate/diversity controls or empty execution results. Only 2 of 4,925 candidates were schema-invalid after the Cypher checks. A rewrite audit found that all reported-run candidates were already identical to their normalized Cypher, so accepted examples do not depend on semantics-changing rewrites. The ablations should therefore be read as a reliability study of the execution-grounded core: in the target-100 suite, every non-unconstrained graph/setting cell reached all eight category targets, while unconstrained generation did not produce balanced executable coverage. Across the three evidence-ready suites, target-normalized coverage is 1.000 for ev-

ery non-unconstrained cell.

**Judge calibration.** An 80-row post-hoc human audit sampled accepted and rejected candidates across both graphs and all categories. Agreement is 80.0%, Cohen’s  $\kappa = 0.60$ , judge precision/specificity are 1.00, recall is 0.714, and no false accepts appear in the sample. The judge is conservative and protects accepted-example quality; human labels calibrate the gate but do not participate in generation.

**RQ3: downstream stress test.** We evaluate downstream Text2Cypher by giving each model the schema text and the question, then scoring the generated query by live execution on FinBench or SNB. This is where the benchmark becomes useful: many outputs parse, mention plausible schema, and still answer the wrong graph question. The local Qwen3.5-9B baseline, for example, reaches 0.963 parse validity and 0.916 schema validity but only 0.189 exact execution accuracy on the 296-example held-out split. In an 11-model completed local transfer suite, zero-shot execution accuracy ranges from 0.000 to 0.203 (mean 0.036). Table 5 gives the control comparison. The primary few-shot generalization result is the scored no-signature control, which excludes exact query-signature matches and near-duplicate questions and raises mean accuracy to 0.200. Ordered and random same-category example banks reach 0.269/0.267, but we treat them as operational upper bounds because they often share query signatures. This split is important for industry use: the benchmark can be a hard model-evaluation set and, separately, a private example bank for schema-specific retrieval or adaptation.

| Condition    | Sig.  | Mean  | Best  |
|--------------|-------|-------|-------|
| Zero-shot    | –     | 0.036 | 0.203 |
| No-signature | 0.000 | 0.200 | 0.828 |
| Ordered      | 0.866 | 0.269 | 0.993 |
| Random       | 0.854 | 0.267 | 0.986 |

Table 5: Eleven-model local downstream transfer controls on the 296-example held-out split. Sig. is the fraction of selected demonstrations sharing the test query signature; the no-signature row is the leakage-aware example-bank result, while ordered and random same-category rows are operational upper bounds.

**RQ4: third-graph onboarding.** ICIJ onboarding reaches 800 accepted examples from 983 candidates on a 2.0M-node, 3.3M-edge public finance/compliance graph, with 100 examples in every category. This is evidence from a third public



graph, not proof of private-tenant coverage. It is still important because it exercises schema-derived relationship-count, anti-join, and top-k templates beyond the two LDBC workloads.

## 7 Industry Use

PIPE-Cypher is meant to be rerun when an enterprise graph changes. Each example records the schema snapshot, graph profile, model identifier, validation gates, execution sample, judge scores, difficulty features, and source run. Teams can refresh after schema changes, compare models on the same held-out split, inspect failures by category, build a schema-specific question-answer bank for retrieval or adaptation, and export redacted review packets. A deployment needs read-only graph credentials, schema introspection, a bounded value policy, a local endpoint, a dry run, scaled generation, judge calibration, and redacted export. We validate this pattern on FinBench, SNB, and ICIJ while keeping the private-tenant gap explicit.

## 8 Conclusion

PIPE-Cypher reframes Text2Cypher benchmarking as a private, repeatable enterprise workflow. The main lesson is that generation improves when Cypher constraints become executable checks. Reverse grounding makes questions answerable. Deterministic validation catches unsafe or schema-invalid queries. Execution exposes empty or brittle candidates. Diversity diagnostics reveal concentration. A calibrated local judge adds a conservative semantic filter. Together these pieces produce a benchmark that is balanced, auditable, refreshable, and able to reveal downstream model failures that syntax-only evaluation would hide.

## Limitations

Execution validity does not guarantee semantic correctness. The completed 80-row, single-audit-sheet calibration suggests a conservative judge with no observed false accepts in the labeled sample, but the confidence interval is wider than the point estimate and larger multi-annotator audits may reveal additional failure modes. FinBench, SNB, and ICIJ are public enterprise-style proxies rather than a proprietary tenant graph, so we test the onboarding pattern but not every deployment constraint of a real organization. The full export is balanced by graph, category, and difficulty, but

query-signature diagnostics still show template concentration from seeded, execution-grounded generation. PIPE-Cypher deliberately disallows or skips risky Cypher constructs such as writes, undirected relationships, UNION, CALL, UNWIND, and parser-risky rewrites. This is a safe benchmark-generation subset, not complete coverage of every production Cypher idiom. The downstream few-shot result should be read as graph-specific example-bank conditioning: the no-signature control is the leakage-aware result, while ordered and random same-category demonstrations are upper-bound conditions that often share query signatures. Tenant-specific fine-tuning remains an engineering path enabled by the artifact, not a completed deployment claim here. The redaction audit checks exact residuals for known value-bearing strings, but it is not a full PII classifier and does not make schema names confidential.

## Ethics Statement

PIPE-Cypher is designed for private benchmark generation under local-model deployment constraints. Organizations using it should restrict benchmark artifacts to authorized users, review sampled values for sensitive content, and document whether judge calibration relied on human annotators.

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## A Prior-Work Positioning

Table 6 expands the related-work comparison from the main paper. PIPE-Cypher’s contribution is the way answerability, governance, privacy, judging, and refresh are assembled into an organization-run benchmark factory.

| Prior mechanism                                        | PIPE-Cypher delta                                                                                  |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Template/LLM synthesis in SyntheT2C and Auto-Cypher    | Outcome-aware reverse grounding binds slots through live Cypher before NL realization.             |
| Execution validation in Auto-Cypher and Mind the Query | Execution is one gate in a ledger with direction, read-only, value, non-empty, and judge evidence. |
| Schema/value checks in Mind the Query                  | Checks are packaged for private refresh with configurable value sampling and redacted exports.     |
| Human logical review in Mind the Query                 | Human labels calibrate a local LLM judge; they are not a generation gate.                          |
| Public static datasets and tuning corpora              | Organization-run benchmark factory with manifests, refresh, and provenance audits.                 |

Table 6: Prior mechanism comparison. PIPE-Cypher combines outcome-aware grounding, deterministic Cypher checks, local judge calibration, privacy policy, refresh support, and audit logs so organizations can generate their own benchmarks rather than only consume static datasets.

## B Experimental Setup and Exported Benchmark

Unless noted otherwise, the extended results use the same live FinBench/SNB export as the main results. Tables in this section pin down the scale, graph mix, split, and validation totals so the downstream and diversity analyses are tied to a single benchmark artifact.

### B.1 Benchmark Artifact Summary

Tables 7, 8, and 9 give the run configuration, export manifest summary, and gate/distribution details for the 3,000-example benchmark.

| Setting                | Value                               |
|------------------------|-------------------------------------|
| Accepted examples      | 3,000                               |
| Primary graph          | LDBC FinBench, 2,000 examples       |
| Secondary graph        | LDBC SNB, 1,000 examples            |
| Categories             | 8 balanced categories, 375 each     |
| Generation/judge model | Local Qwen3.5-9B                    |
| Execution backend      | Neo4j Community, two live databases |
| Judge audit packet     | 80 sampled accepted/rejected pairs  |

Table 7: Full live experimental setup used for the generated benchmark artifact.

| Export artifact     | Examples | FinBench | SNB   | Train | Dev | Test |
|---------------------|----------|----------|-------|-------|-----|------|
| Live full benchmark | 3,000    | 2,000    | 1,000 | 2,408 | 296 | 296  |

Table 8: Accepted live full benchmark package with stable IDs, gate metadata, result samples, statistics, and manifest hash `cf274344be2abe7e`.

| Artifact property                   | Value                     |
|-------------------------------------|---------------------------|
| Categories                          | 8 balanced categories     |
| FinBench/category                   | 250                       |
| SNB/category                        | 125                       |
| Difficulty split                    | 1,569 easy / 1,431 medium |
| Used labels / rel. types            | 16 / 17                   |
| Read/syntax/schema/exec/judge gates | 3,000/3,000               |

Table 9: Distribution and gate summary for the exported full benchmark artifact.

## C Governed Generation Evidence

The central reliability question is whether reverse grounding and Cypher validation can fill every planned graph/category cell at target scale.

### C.1 Target-100 Stress Baseline

Figure 2 and Table 10 show the target-100 stress baseline. The unconstrained rows are deliberately harsh: raw local-model generations may look plausible, but they do not produce balanced, executable benchmark coverage. The governed variants fill every target cell on both graphs.

| Setting           | Graph    | Attempts | Records | Accepted | Acceptance | Categories at target |
|-------------------|----------|----------|---------|----------|------------|----------------------|
| Unconstrained LLM | FinBench | 422      | 422     | 200      | 0.474      | 2/8                  |
| Unconstrained LLM | SNB      | 2,000    | 2,000   | 50       | 0.025      | 0/8                  |
| Reverse-only      | FinBench | 815      | 815     | 800      | 0.982      | 8/8                  |
| Reverse-only      | SNB      | 820      | 820     | 800      | 0.976      | 8/8                  |
| Validators+repair | FinBench | 833      | 833     | 800      | 0.960      | 8/8                  |
| Validators+repair | SNB      | 824      | 824     | 800      | 0.971      | 8/8                  |
| No retrieval      | FinBench | 819      | 819     | 800      | 0.977      | 8/8                  |
| No retrieval      | SNB      | 809      | 809     | 800      | 0.989      | 8/8                  |
| No rewrite        | FinBench | 826      | 826     | 800      | 0.969      | 8/8                  |
| No rewrite        | SNB      | 834      | 834     | 800      | 0.959      | 8/8                  |
| No LLM judge      | FinBench | 812      | 812     | 800      | 0.985      | 8/8                  |
| No LLM judge      | SNB      | 828      | 828     | 800      | 0.966      | 8/8                  |
| Full PIPE-Cypher  | FinBench | 824      | 824     | 800      | 0.971      | 8/8                  |
| Full PIPE-Cypher  | SNB      | 824      | 824     | 800      | 0.971      | 8/8                  |

Table 10: Live target-100 ablation evidence with local Qwen3.5-9B. Governed graph runs target 100 accepted examples per category; the unconstrained row is a stress baseline reported with explicit attempt accounting.

### C.2 Gate-Level Quality

Yield alone is not enough. A benchmark factory must show which checks keep bad examples out. Table 11 and Figure 3 report read-only, syntax, schema, execution, and judge/post-hoc rates for the same target-100 suite. Read-only and syntax checks saturate once governance is active; execution and semantic judging expose the remaining differences.

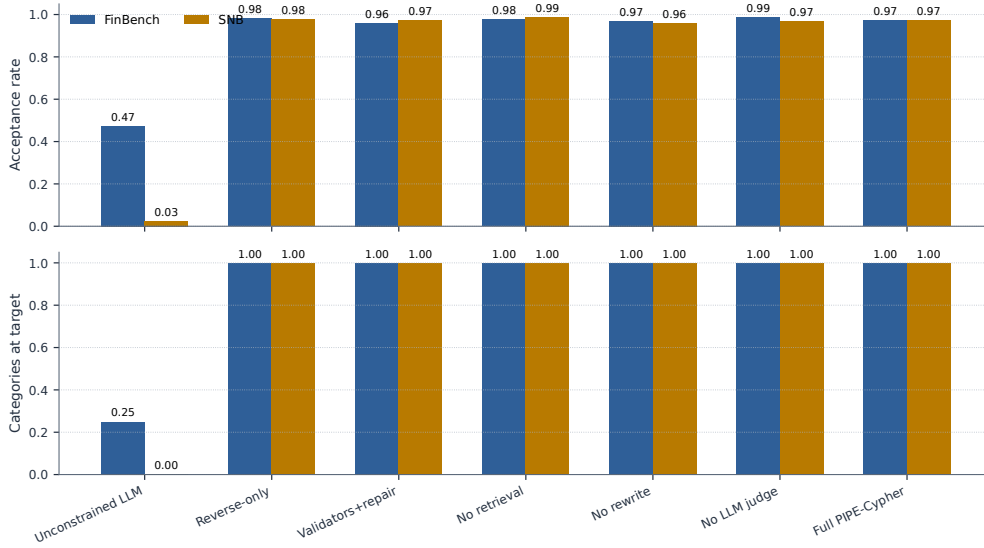


Figure 2: Target-100 ablation yield on FinBench and SNB. Unconstrained local generation is reported as a stress baseline with explicit attempt accounting; the execution-grounded governed variants reach all eight workload-category targets on both graphs. The result supports reliability of the grounded governance core without implying that every optional module is required for yield.

| Setting           | Graph    | Read-only | Syntax | Schema | Exec. | Judge/post-hoc |
|-------------------|----------|-----------|--------|--------|-------|----------------|
| Unconstrained LLM | FinBench | 1.000     | 1.000  | 0.806  | 0.723 | 0.557          |
| Unconstrained LLM | SNB      | 1.000     | 0.998  | 0.599  | 0.478 | 0.144          |
| Reverse-only      | FinBench | 1.000     | 1.000  | 0.982  | 0.982 | 0.982          |
| Reverse-only      | SNB      | 1.000     | 1.000  | 0.998  | 0.998 | 0.976          |
| Validators+repair | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.960          |
| Validators+repair | SNB      | 1.000     | 1.000  | 0.998  | 0.998 | 0.971          |
| No retrieval      | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.977          |
| No retrieval      | SNB      | 1.000     | 1.000  | 0.998  | 0.998 | 0.989          |
| No rewrite        | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.969          |
| No rewrite        | SNB      | 1.000     | 1.000  | 0.998  | 0.998 | 0.959          |
| No LLM judge      | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.985          |
| No LLM judge      | SNB      | 1.000     | 1.000  | 0.998  | 0.998 | 0.966          |
| Full PIPE-Cypher  | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.971          |
| Full PIPE-Cypher  | SNB      | 1.000     | 1.000  | 0.998  | 0.998 | 0.971          |

Table 11: Quality-gate rates for the live target-100 ablation suite. Rates are computed over all generated records in each graph/setting; for no-judge settings, the judge column is a post-hoc scoring diagnostic.



Figure 3: Gate-rate heatmap for the audited target-100 ablation suite. Deterministic read-only and syntax gates are saturated; execution and judge rates expose the remaining quality differences across graph and pipeline variants.

### C.3 Repeated-Suite Stability

The same pattern holds across target sizes and seeds. The repeated-suite comparison normalizes by each suite’s planned target, so it measures stability rather than rewarding larger raw counts. Full PIPE-Cypher and the governed ablations reach complete target coverage on both graphs, which is the behavior we want from a repeatable benchmark generator.

| Setting           | Graph    | Suites | Target cov. | Acceptance  | Cat. target | Exec. | Judge |
|-------------------|----------|--------|-------------|-------------|-------------|-------|-------|
| No LLM judge      | FinBench | 3/3    | 1.000       | 0.994±0.008 | 1.000       | 1.000 | 0.994 |
| No retrieval      | FinBench | 3/3    | 1.000       | 0.967±0.031 | 1.000       | 1.000 | 0.967 |
| No rewrite        | FinBench | 3/3    | 1.000       | 0.969±0.023 | 1.000       | 1.000 | 0.969 |
| Full PIPE-Cypher  | FinBench | 3/3    | 1.000       | 0.975±0.016 | 1.000       | 1.000 | 0.975 |
| Reverse-only      | FinBench | 3/3    | 1.000       | 0.994±0.011 | 1.000       | 0.994 | 0.994 |
| Validators+repair | FinBench | 3/3    | 1.000       | 0.987±0.023 | 1.000       | 1.000 | 0.987 |
| No LLM judge      | SNB      | 3/3    | 1.000       | 0.982±0.015 | 1.000       | 0.996 | 0.982 |
| No retrieval      | SNB      | 3/3    | 1.000       | 0.993±0.004 | 1.000       | 0.996 | 0.993 |
| No rewrite        | SNB      | 3/3    | 1.000       | 0.983±0.021 | 1.000       | 0.996 | 0.983 |
| Full PIPE-Cypher  | SNB      | 3/3    | 1.000       | 0.987±0.014 | 1.000       | 0.996 | 0.987 |
| Reverse-only      | SNB      | 3/3    | 1.000       | 0.989±0.011 | 1.000       | 0.996 | 0.989 |
| Validators+repair | SNB      | 3/3    | 1.000       | 0.987±0.014 | 1.000       | 0.996 | 0.987 |

Table 12: Target-size and repeated-seed ablation sensitivity. Target coverage normalizes accepted examples by each suite’s planned graph/category target, so target-50 and target-100 suites can be compared without treating larger raw counts as quality gains. Unconstrained local generation is excluded from this stability table and reported separately as the attempt-logged stress baseline in Table 10.

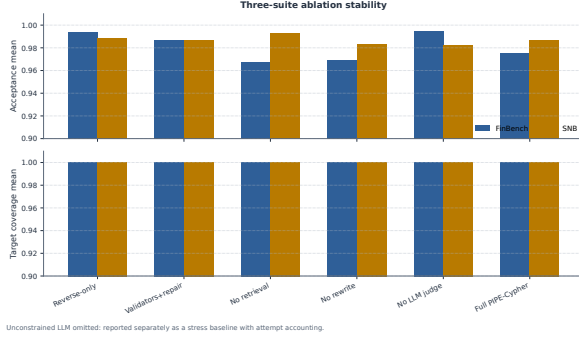


Figure 4: Three-suite ablation stability over the original target-50 suite, corrected target-100 suite, and seed-17 target-50 repeat. Target-normalized coverage stays at 1.000 for all non-unconstrained cells.

#### C.4 Rejected Candidate Taxonomy

The rejection taxonomy shows where the remaining work occurs. After Cypher validation, schema-invalid candidates are rare. Most rejections are duplicate/diversity blocks from recovery or empty-result candidates caught before export. This is the behavior an enterprise deployment wants: invalid or brittle examples remain in the ledger, not in the benchmark.

| Failure bucket              | Count | Share of rejected |
|-----------------------------|-------|-------------------|
| Diversity/duplicate control | 1,366 | 0.710             |
| Empty result                | 486   | 0.252             |
| Judge semantic reject       | 67    | 0.035             |
| Execution error             | 4     | 0.002             |
| Schema invalid              | 2     | 0.001             |

Table 13: Failure taxonomy over full-run generation candidates before benchmark export. Accepted examples are excluded from the bucket shares.

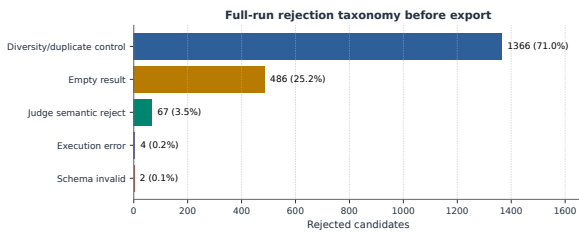


Figure 5: Full-run rejection taxonomy before benchmark export. Most rejected candidates come from duplicate/diversity control during category recovery and from empty execution results; schema-invalid Cypher is rare after Cypher validation.

## D Benchmark Artifact and Third-Graph Onboarding

### D.1 Export Balance

The exported benchmark makes scale and balance visible. Figure 6 shows the planned 2:1 FinBench/SNB mix, exact category balance, and near-even difficulty split. PIPE-Cypher produces a benchmark that is balanced by design rather than sampled opportunistically.

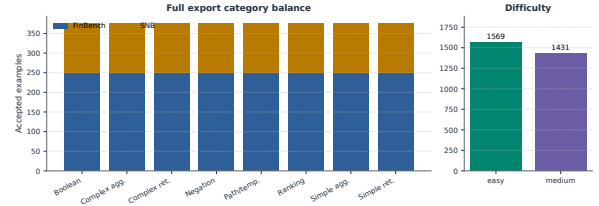


Figure 6: Full 3,000-example export distribution. The benchmark preserves the planned 2:1 FinBench/SNB graph mix while balancing all eight Cypher workload categories and maintaining a near-even easy/medium difficulty split.

### D.2 Graph Scale and Schema Variety

Table 14 summarizes the graphs before the third-graph onboarding audit. FinBench and SNB are the controlled LDBC workloads. ICIJ is the public finance/compliance graph we use to test whether the same onboarding code works outside the two original schemas.

| Graph               | Nodes     | Relationships | Labels | Rel. types | Study status     |
|---------------------|-----------|---------------|--------|------------|------------------|
| FinBench            | 10,006    | 57,622        | 5      | 9          | reported         |
| SNB                 | 34,735    | 70,842        | 14     | 15         | reported         |
| ICIJ Offshore Leaks | 2,016,523 | 3,339,267     | 5      | 14         | onboarding audit |

Table 14: Study graph statistics. ICIJ Offshore Leaks is used as a public third-graph onboarding audit for arbitrary finance/compliance schemas beyond the two LDBC study workloads.

### D.3 ICIJ Onboarding

Table 15 and Figure 7 report the third-graph result. ICIJ matters because it forced PIPE-Cypher to derive sparse-category templates from the schema instead of relying on preauthored FinBench/SNB templates. The run reaches all eight category targets while using the same validation and audit standards.

| ICIJ onboarding property      | Value                                    |
|-------------------------------|------------------------------------------|
| Graph nodes / relationships   | 2,016,523 / 3,339,267                    |
| Labels / relationship types   | 5 / 14                                   |
| Generated / accepted          | 983 / 800                                |
| Acceptance rate               | 0.814                                    |
| Categories at target          | 8/8                                      |
| Study audit                   | ready                                    |
| Sparse schema-derived accepts | complex agg. 97, negation 28, ranking 98 |

Table 15: ICIJ Offshore Leaks third-graph onboarding audit. The public finance/compliance graph tests arbitrary-schema generation beyond the two LDBC study workloads; raw values remain outside the reported artifacts.

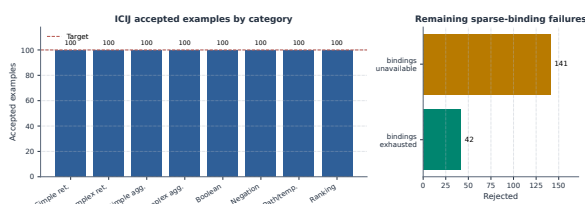


Figure 7: ICIJ Offshore Leaks onboarding audit. Schema-derived sparse-category templates recover balanced target-100 coverage on a public finance/compliance graph beyond the two LDBC workloads.

## D.4 Category Crosswalk

Table 16 connects PIPE-Cypher’s eight categories to the simpler retrieval/aggregation/evaluation-query taxonomy used in Mind the Query. The crosswalk keeps the comparison clear while making our extra enterprise workload classes explicit: negation, temporal/path reasoning, and ranking/top-k.

| PIPE-Cypher category      | Closest MTQ category | Added enterprise distinction                                 |
|---------------------------|----------------------|--------------------------------------------------------------|
| Simple retrieval          | SR                   | Direct node/edge lookup with exact filters.                  |
| Complex retrieval         | CR                   | Multi-hop or multi-pattern retrieval.                        |
| Simple aggregation        | SA                   | Single aggregation such as count/min/max/average.            |
| Complex aggregation       | CA                   | Grouped or multi-stage aggregation over graph neighborhoods. |
| Boolean existence         | EQ                   | Precise yes/no or existence answer.                          |
| Negation/difference       | CR                   | Absence, anti-join, or difference query.                     |
| Path/temporal transaction | CR/CA                | Temporal or path-oriented transaction neighborhood.          |
| Ranking/top-k             | SA/CA                | Ordered top-k query with explicit limit.                     |

Table 16: Category crosswalk to Mind the Query. PIPE-Cypher keeps the familiar retrieval/aggregation/evaluation-query structure while adding enterprise workloads such as negation, temporal paths, and ranking.

## E Downstream Transfer and Example-Bank Utility

### E.1 Transfer Controls

The downstream experiment tests whether the benchmark is useful for model evaluation. A useful enterprise benchmark should not merely reward syntactically valid Cypher. It should reveal when a model produces a query that parses, mentions plausible schema elements, and even executes,

but answers the wrong operational question. The Qwen3.5-9B result has exactly that shape: parse validity is 0.963 and schema validity is 0.916, while exact execution accuracy is only 0.189 on the full held-out split. The gap shows that PIPE-Cypher measures semantic graph-query competence rather than only query formatting.

The multi-model transfer suite compares local general instruction, code-tuned, Cypher instruction, and Text2Cypher-finetuned models while keeping the schema prompt, evaluation backend, split, and execution metrics fixed. Figure 8 shows a sharp pattern: zero-shot transfer is weak, but accepted graph-specific examples can become a useful private question-answer bank. Demonstration-bank controls close much of the gap for compatible model families such as Qwen, Qwen-Coder, and one Gemma Text2Cypher LoRA. Several public finetuned checkpoints remain brittle under enterprise-style prompting.

Table 18 reports checkpoint-level uncertainty for the same controls. The interval is deliberately conservative because the unit is model checkpoint rather than question row; it supports the narrower claim that example-bank gains are model-family dependent, not universal.

| Control                   | Mean acc. | $\Delta$ vs zero | 95% $\Delta$ CI | Models improved |
|---------------------------|-----------|------------------|-----------------|-----------------|
| Ordered same-category     | 0.269     | 0.233            | [0.000, 0.481]  | 3/11            |
| Scored no-signature       | 0.200     | 0.163            | [0.000, 0.335]  | 3/11            |
| Random same-category mean | 0.267     | 0.231            | [0.000, 0.464]  | 3/11            |

Table 18: Model-level paired bootstrap uncertainty for downstream few-shot controls. The unit of resampling is the local checkpoint, not an individual question, so the interval is a conservative check on whether gains are broad across model families. Zero-shot mean execution accuracy is 0.036.

Table 19 gives the interpretation for the transfer result. Ordered and random same-category demonstrations are useful example-bank conditions. The scored no-signature condition is the stricter leakage-aware control.

| Selection mode | Rows | Demos | Sig. match | High sim. | Mean sim. |
|----------------|------|-------|------------|-----------|-----------|
| Ordered        | 296  | 1480  | 0.866      | 0.389     | 0.825     |
| No-signature   | 296  | 1480  | 0.000      | 0.000     | 0.585     |
| Random         | 296  | 1480  | 0.854      | 0.409     | 0.824     |

Table 19: Few-shot leakage controls for downstream demonstration-bank evaluation. The held-out split has 0 exact train/test question overlaps and 289 train/test query-signature overlaps; selection-mode rates show how often retrieved demonstrations share the test query signature or exceed the 0.90 normalized-question similarity threshold.

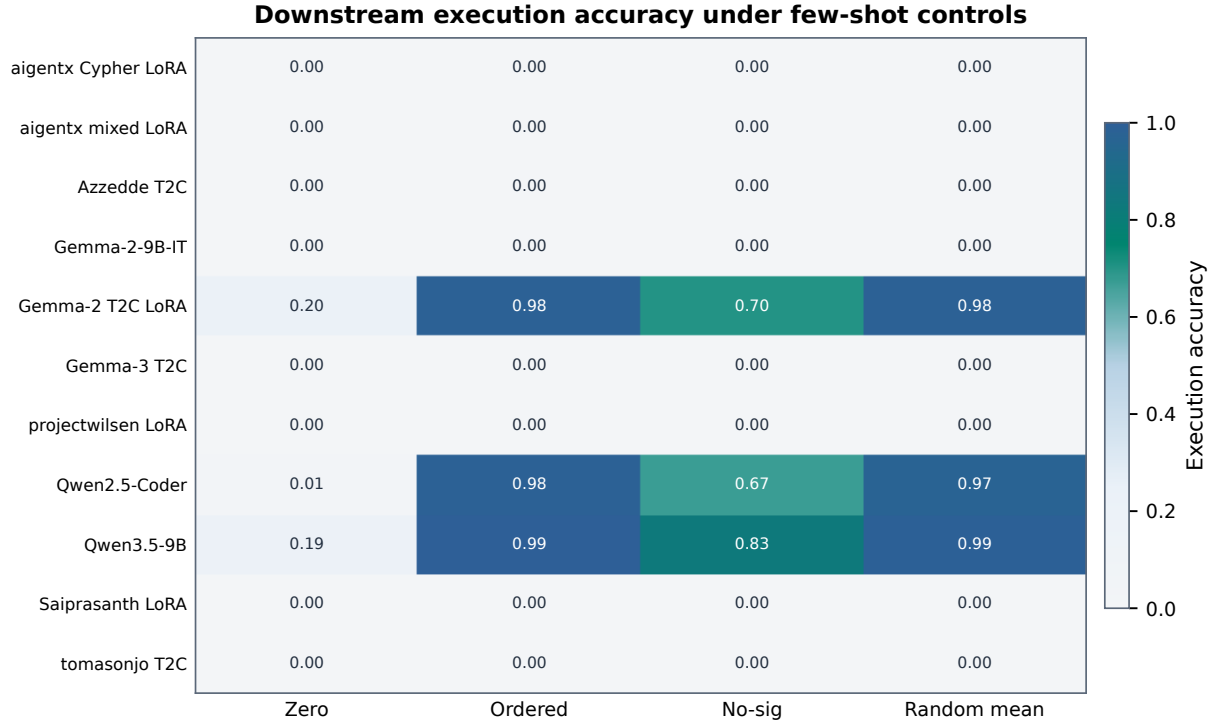


Figure 8: Execution accuracy for 11 completed local downstream models under zero-shot and few-shot control modes. The heatmap highlights both findings: schema-specific examples can sharply help compatible models, and several public fine-tuned checkpoints still fail under enterprise-style Cypher prompting.

| Model                                       | Tuning                 | Zero  | Ordered | No-sig | Random $\mu$ | Random $\sigma$ | Best            |
|---------------------------------------------|------------------------|-------|---------|--------|--------------|-----------------|-----------------|
| aigentx/Llama-3.1-8B Cypher LoRA            | Cypher LoRA            | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |
| aigentx/Llama-3.1-8B Cypher mixed LoRA      | Cypher mixed LoRA      | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |
| Azzedde/Llama3.1-8b-text2cypher             | Text2Cypher fine-tuned | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |
| Gemma-2-9B-IT                               | general instruction    | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |
| neo4j/Gemma-2-9B Text2Cypher LoRA           | Text2Cypher LoRA       | 0.203 | 0.983   | 0.699  | 0.981        | 0.009           | ordered (0.983) |
| neo4j/Gemma-3-4B Text2Cypher                | Text2Cypher fine-tuned | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |
| projectwilsen/Llama-3.1-8B Text2Cypher LoRA | Text2Cypher LoRA       | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |
| Qwen2.5-Coder-7B-Instruct                   | code instruction       | 0.007 | 0.983   | 0.669  | 0.972        | 0.002           | ordered (0.983) |
| Qwen3.5-9B                                  | general instruction    | 0.189 | 0.993   | 0.828  | 0.986        | 0.007           | ordered (0.993) |
| Saiprasanth15/Llama-3.1-8B Text2Cypher LoRA | Text2Cypher LoRA       | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |
| tomasonjo/text2cypher-demo-16bit            | Text2Cypher fine-tuned | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |

Table 17: Few-shot demonstration-bank controls for local downstream Text2Cypher evaluation over completed 296-example local-model runs. Ordered uses the deterministic same-graph, same-category example bank; scored excludes exact query-signature matches and near-duplicate questions; random reports the mean and standard deviation across seeds 13, 17, and 23. “No gain” means no few-shot control exceeded that model’s zero-shot execution accuracy.



## E.2 Downstream Failure Modes

The failure taxonomy appears next to the transfer results because it explains why the benchmark is useful operationally. Incorrect rows are not all the same. Some outputs fail to parse. Some mention invalid schema. Some fail at execution time. Many execute but answer the wrong question. That separation tells a benchmark owner whether to improve schema retrieval, prompt constraints, value grounding, or tenant-specific adaptation.

| Downstream outcome | Count | Share of incorrect |
|--------------------|-------|--------------------|
| Answer mismatch    | 125   | 0.521              |
| Execution failed   | 79    | 0.329              |
| Schema invalid     | 25    | 0.104              |
| Parse invalid      | 11    | 0.046              |

Table 20: Downstream Text2Cypher failure taxonomy for local Qwen3.5-9B on the full exported test split. Shares exclude exact-answer matches.

## E.3 Supplementary Text Metrics

Table 21 lists optional text-overlap metrics for debugging answer rendering and near-match behavior. These are not correctness metrics; the reported correctness results use execution accuracy and answer-set F1.

| Metric       | Supported use and primary citation                                                 |
|--------------|------------------------------------------------------------------------------------|
| ROUGE-1/2/L  | Reference overlap over serialized answer sets and query strings (Lin, 2004)        |
| BLEU         | Sentence-level reference overlap with brevity penalty (Papineni et al., 2002)      |
| METEOR       | Exact-token precision/recall with fragmentation penalty (Banerjee and Lavie, 2005) |
| BERTScore    | Optional contextual-embedding similarity (Zhang et al., 2020)                      |
| FrugalScore  | Optional efficient learned NLG metric (Kamal Eddine et al., 2022)                  |
| Cosine       | Token-count vector cosine similarity (Manning et al., 2008)                        |
| Jaro-Winkler | Character-level near-match similarity (Jaro, 1989; Winkler, 1990)                  |
| Exact match  | Raw and normalized string exact match (Rajpurkar et al., 2016)                     |

Table 21: Supplementary reference-based text metrics supported by the PIPE-Cypher evaluator. These metrics are useful for debugging answer rendering, paraphrase sensitivity, and near-match behavior, but they do not replace execution accuracy or answer-set F1 for Text2Cypher correctness. BERTScore and FrugalScore are optional integrations because they require additional metric/model packages.

## F Diversity and Cypher Strategy Audit

### F.1 Diversity Diagnostics

Aggregate acceptance rates hide too much. The diversity and strategy diagnostics show whether the benchmark is balanced, which Cypher operators it exercises, and where residual concentration remains. Figure 9 shows the useful diversity picture: category, graph-category, and difficulty balance are strong by construction, while schema and value coverage remain quantities to monitor during refresh.

| Metric                                  | Value                 |
|-----------------------------------------|-----------------------|
| PIPE-Diversity index                    | 0.549                 |
| Question Distinct-1                     | 0.039                 |
| Question Distinct-2                     | 0.085                 |
| Question adjusted Distinct-2            | 0.266                 |
| Question self-BLEU-2 (sampled)          | 0.813                 |
| Mean nearest-neighbor question Jaccard  | 0.775                 |
| Unique query-signature ratio            | 0.041                 |
| Top query-signature share               | 0.083                 |
| Template-family entropy                 | 0.826                 |
| Operator-combination entropy            | 0.880                 |
| Unique structural substructures         | 134                   |
| Category normalized entropy             | 1.000                 |
| Graph-category normalized entropy       | 0.980                 |
| Difficulty normalized entropy           | 0.998                 |
| Label coverage                          | 0.941                 |
| Relationship-type coverage              | 0.708                 |
| Property-name coverage                  | 0.426                 |
| Unique grounded-value ratio             | 0.357                 |
| Grounded values exactly quoted          | 0.823                 |
| Aggregation / negation / ordering rates | 0.500 / 0.125 / 0.125 |

Table 22: Diversity diagnostics for the full exported benchmark. PIPE-Diversity is a geometric mean of lexical, query-template, structural, schema, value, and balance components; component rows are shown so the composite score does not hide residual concentration.

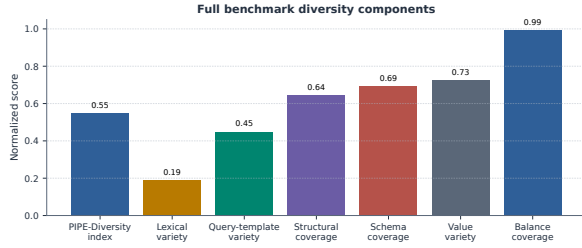


Figure 9: Diversity diagnostics for the full 3,000-example export. Category, graph-category, and difficulty balance are strong by construction; schema coverage and query-signature diversity expose remaining concentration from the seeded-template run.

## F.2 Diversity-Governed Selection

Table 23 reports the first diversity-improvement pass. We select a target-50-per-graph/category subset from the 3,000 accepted examples and compare it with a hash-balanced random subset of the same size. The selector uses an MMR-style novelty score over query signatures, template families, structural substructures, schema atoms, entity values, and question tokens after all quality gates have passed. It improves the PIPE-Diversity index, unique query-signature ratio, unique structural substructures, adjusted Distinct-2, and property coverage while preserving a signature-disjoint 640/80/80 split. The mixed template-family entropy and self-BLEU result is useful rather than embarrassing: it shows that post-hoc selection can improve coverage, but stronger diversity requires oversampling or inducing more source templates during generation when a graph/category cell has only one or two viable signatures.

| Metric                              | Random balanced | Diversity governed | $\Delta$ |
|-------------------------------------|-----------------|--------------------|----------|
| PIPE-Diversity index                | 0.557           | 0.575              | +0.017   |
| Unique query-signature ratio        | 0.062           | 0.135              | +0.073   |
| Top signature share (lower better)  | 0.062           | 0.062              | +0.000   |
| Template-family entropy             | 0.903           | 0.868              | -0.035   |
| Operator-combination entropy        | 0.901           | 0.911              | +0.010   |
| Unique structural substructures     | 97.000          | 134.000            | +37.000  |
| Question self-BLEU-2 (lower better) | 0.850           | 0.866              | +0.016   |
| Adjusted Distinct-2                 | 0.266           | 0.284              | +0.018   |
| Property coverage                   | 0.407           | 0.426              | +0.019   |

Table 23: Balanced subset comparison at the same graph/category target. The diversity-governed selector applies MMR-style novelty over Cypher signatures, template families, structural substructures, schema atoms, values, and question tokens after quality gates have already passed; structural/schema gains are reported alongside residual template concentration.

## F.3 Cypher Strategy Coverage

Strategy diagnostics complement category balance. Two questions can share a category while exercising different Cypher behavior. Conversely,

a category-balanced dataset can still miss joins, paths, negation, ranking, or bounded-result patterns. The strategy matrix and downstream strategy outcomes show what the benchmark actually exercises.

| Primary strategy | Examples | Share | Rel. patterns | Return arity | Downstream exec. acc. |
|------------------|----------|-------|---------------|--------------|-----------------------|
| Aggregation      | 1125     | 0.375 | 1.42          | 1.00         | 0.396                 |
| Join-heavy       | 375      | 0.125 | 2.33          | 2.33         | 0.000                 |
| Negation         | 375      | 0.125 | 1.89          | 2.84         | 0.000                 |
| Order/rank       | 375      | 0.125 | 1.87          | 3.41         | 0.000                 |
| Path             | 375      | 0.125 | 1.37          | 3.00         | 0.000                 |
| Single hop       | 375      | 0.125 | 1.00          | 2.33         | 0.324                 |

Table 24: Cypher strategy diagnostics over the full 3,000-example export. Strategy tags are derived from generated Cypher structure rather than from category labels; downstream execution accuracy is reported on the full held-out test split when a strategy appears there.

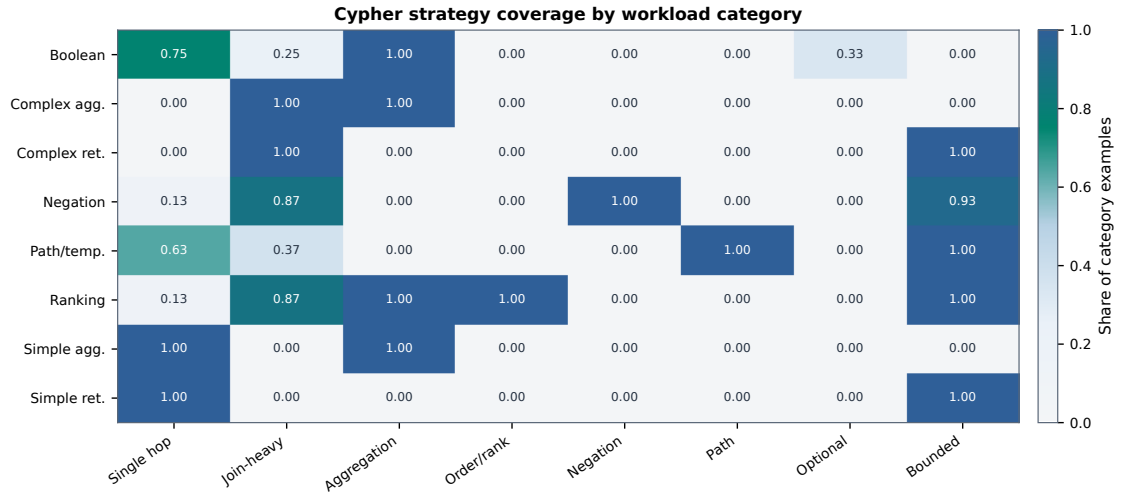


Figure 10: Cypher strategy coverage by workload category. Category balancing does not by itself guarantee operator coverage; the strategy matrix shows where the benchmark exercises single-hop retrieval, joins, aggregation, ordering, negation, path patterns, optional matches, and bounded-result queries.

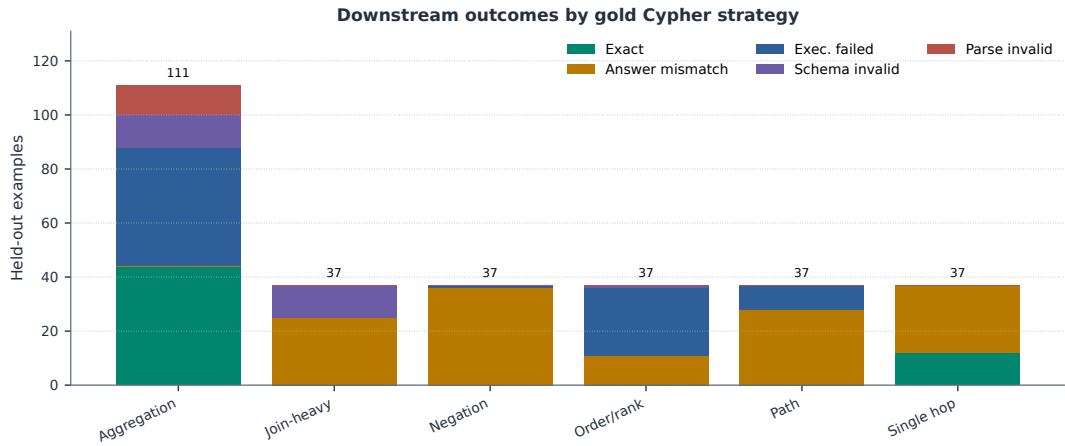


Figure 11: Downstream outcomes by gold Cypher strategy on the full held-out test split. The local Qwen3.5-9B baseline fails differently across strategies: aggregation mixes exact matches with execution failures, while join-heavy, negation, path, and ranking examples are dominated by semantically wrong executable queries or execution failures.

## G Governance, Judge Calibration, and Deployment Details

The deployment details below matter because enterprise failures rarely come from a single model call. They come from missing safety boundaries, unclear value policies, weak audit trails, or no explanation for why an example was accepted. The validator cascade, prompt contracts, automation comparison, and judge audit describe the controls that make PIPE-Cypher a governed local workflow rather than a manual dataset-writing exercise.

### G.1 Validation Cascade

Table 25 lists the deterministic and judge gates in execution order. It shows how an enterprise deployment keeps unsafe, schema-invalid, empty, or semantically weak examples out of exported benchmarks.

| Gate or ledger             | Count | Denominator |
|----------------------------|-------|-------------|
| Export accepted            | 3,000 | 3,000       |
| Read-only safety           | 3,000 | 3,000       |
| Syntax validity            | 3,000 | 3,000       |
| Schema/value validity      | 3,000 | 3,000       |
| Execution success          | 3,000 | 3,000       |
| LLM judge pass             | 3,000 | 3,000       |
| Rejected candidates logged | 1,925 | 4,925       |

Table 25: PIPE-Cypher validation cascade for the full export and its logged candidate ledger. Unlike Mind the Query, human review is calibration-only.

### G.2 Rewrite and Governance Audits

Table 26 addresses rewrite preservation directly. In the reported generation records, generated Cypher was already identical to normalized Cypher, so the accepted benchmark does not rely on a semantics-changing RETURN DISTINCT insertion or projection rewrite. Rewriting is still part of the pipeline, but in this run its role is conservative validation and logging rather than silent semantic alteration.

| Rewrite audit property              | Value |
|-------------------------------------|-------|
| Generation records audited          | 4,925 |
| Accepted records audited            | 3,000 |
| Records changed by normalization    | 0     |
| Accepted records changed            | 0     |
| RETURN DISTINCT insertions          | 0     |
| Accepted RETURN DISTINCT insertions | 0     |
| Rewrite-skip reasons logged         | 196   |
| Live comparisons required           | 0     |
| Answer-set equality in comparisons  | 0     |

Table 26: Rewrite prevalence and impact audit over reported generation records. When no generated query differs from its normalized form, no live original/normalized re-execution is required for semantic drift.

Table 27 separates direction, schema/value, syntax/parser, and read-only issues across generation, ablation, and downstream prediction artifacts. The full generation records have no direction failures after validation. Downstream predictions still contain direction errors, which is exactly the kind of graph-specific failure a Cypher benchmark should expose.

| Evidence source         | Direction | Schema/value | Syntax/parser | Read-only |
|-------------------------|-----------|--------------|---------------|-----------|
| Full generation records | 0         | 2            | 0             | 0         |
| Target-size ablations   | 260       | 988          | 5             | 0         |
| Downstream predictions  | 25        | 9            | 12            | 0         |
| Combined                | 285       | 999          | 17            | 0         |

Table 27: Governance failure audit. Direction errors, schema/value errors, syntax/parser failures, and read-only violations are counted separately so the appendix shows which Cypher-specific gates do real work.

### G.3 Gate-Impact Counterfactual

Table 28 summarizes the first gate that blocks each non-accepted candidate. This gives the counterfactual view missing from a yield-only ablation: it shows what kind of bad example would enter the benchmark if a deployment weakened duplicate/diversity controls, non-empty execution, judge review, schema validation, direction checking, or read-only safety.

| First blocking gate | Candidates | Share |
|---------------------|------------|-------|
| Accepted            | 3,000      | 0.609 |
| Duplicate/diversity | 1,366      | 0.277 |
| Empty result        | 486        | 0.099 |
| Judge reject        | 67         | 0.014 |
| Schema              | 2          | 0.000 |
| Execution failure   | 4          | 0.001 |

Table 28: Counterfactual first-blocking-gate audit over generation records. The table shows which failure class would enter the benchmark if that gate were removed or weakened.

## G.4 Privacy and Redaction Audit

Table 29 evaluates the redaction policy rather than merely describing it. The audit builds a sensitive-value set from entity bindings, quoted Cypher literals, reverse-grounding literals, and string-valued execution samples. It then applies the configured redactor and exact-matches the raw values against the redacted question, Cypher, entity, and result fields. This does not replace a tenant’s PII classifier, but it gives reviewers a measurable privacy check for the value-bearing fields PIPE-Cypher itself creates.

| Redaction audit property          | Value  |
|-----------------------------------|--------|
| Examples audited                  | 3,000  |
| Sensitive values checked          | 10,956 |
| Examples with sensitive values    | 2,970  |
| Examples with residual raw values | 0      |
| Residual raw-value matches        | 0      |
| Residual rate per checked value   | 0.000  |
| Unique placeholders               | 3,754  |
| Reused placeholders               | 2,342  |
| Max placeholder frequency         | 337    |

Table 29: Exact-match redaction audit over value-bearing benchmark surfaces. The audit checks entity bindings, quoted Cypher literals, reverse grounding literals, and string-valued result samples after applying the configured redaction policy.

## G.5 Operational Accounting

Table 30 reports local-run accounting from completed generation records. These numbers are for deployment planning: acceptance rate and graph execution latency explain throughput bottlenecks without turning the analysis into a paid-API cost comparison.

| Scope    | Records | Accepted | Acceptance | Exec. p50 ms | Exec. p95 ms |
|----------|---------|----------|------------|--------------|--------------|
| Overall  | 4,925   | 3,000    | 0.609      | 11.995       | 32.832       |
| FinBench | 3,405   | 2,000    | 0.587      | 11.837       | 32.398       |
| SNB      | 1,520   | 1,000    | 0.658      | 12.420       | 38.558       |

Table 30: Operational accounting from completed generation records. These are local-run latency and acceptance diagnostics, not paid-API cost claims.

## G.6 Prompt Profiles and Contracts

Table 31 documents the prompt-profile factors used for ablation planning. The full prompt contracts follow immediately after the table, including the exact LLM-judge system prompt and user prompt template used for reported judge decisions.

| Profile                 | Added constraint                                                | Intended evidence                                     |
|-------------------------|-----------------------------------------------------------------|-------------------------------------------------------|
| Schema-only             | Use only visible schema.                                        | Baseline for schema-grounded prompting.               |
| Instructions            | Exact values, datatype rules, no nested aggregations.           | Targets MTQ-style prompt refinements.                 |
| Examples                | Placeholderized retrieved NL-Cypher pairs.                      | Tests whether examples help without extra governance. |
| Examples + instructions | Few-shot plus explicit rules.                                   | Closest controlled analogue to MTQ Table 5.           |
| Full governed           | Production-derived Cypher hints, AST-safe rewrites, judge gate. | PIPE-Cypher production setting.                       |

Table 31: Prompt profiles implemented for Mind-the-Query-style prompt-factorial evaluation. Results are reported only for completed, audited target-50-or-larger suites.

## H Prompt Contracts

PIPE-Cypher treats prompts as versioned implementation artifacts. The list summarizes the prompt contracts used for generation, repair, judging, and downstream evaluation; hashes fingerprint the full prompt constants in the codebase.

- 1. Template generation.** Stage: Workload proposal. SHA-256: 10b25b.

*Contract.* Schema-only labels, relationships, properties, and categorical values; Realistic enterprise analyst wording; At most two typed slots and JSON-only output

- 2. Reverse binding.** Stage: Graph grounding. SHA-256: 48e949.

*Contract.* Read-only MATCH/WHERE/RETURN DISTINCT/LIMIT only; Slot variables named exactly as requested; Forward relationship directions from the schema

- 3. Cypher generation.** Stage: Candidate query. SHA-256: 61c557.

*Contract.* Only schema-visible constructs and observed directions; RETURN DISTINCT for set returns and exact equality for quoted values; Context columns, categorical hints, placeholderized retrieval, and no writes



4. **Repair.** Stage: Validation feedback. SHA-256: 56c419.

*Contract.* Preserve question intent while fixing validation or execution issues; Keep query read-only and schema-grounded; Return only corrected Cypher

5. **LLM judge.** Stage: Quality gate. SHA-256: 421c7b.

*Contract.* Inputs include question, Cypher, relevant schema excerpt, execution rows, and validation summary; Strict JSON scores for ambiguity, semantic alignment, schema use, and difficulty; Categorical values constrain query literals, not observed result-row values; Pass only useful, unambiguous enterprise benchmark examples

6. **Downstream Text2Cypher.** Stage: Model evaluation. SHA-256: 4c07ff.

*Contract.* Read-only Cypher only; Schema-visible constructs and exact direction preservation; RETURN DISTINCT, count/ranking rules, and no explanations

## H.1 LLM Judge Prompt Used in Reported Runs

This is the complete LLM-judge prompt contract used for all reported judge decisions. The local judge receives a JSON-only system instruction and the following user prompt template after we select the schema elements relevant to the candidate query, sample execution rows, and run deterministic validation. The placeholders are filled with the candidate question, Cypher, relevant schema excerpt, execution rows, and validation summary for each reviewed example.

```
System prompt:
You are an expert benchmark engineer. Return
strict JSON only. No markdown or extra text.

User prompt template:
You are judging whether an NL-to-Cypher
benchmark example is acceptable for an
enterprise benchmark.

Graph schema:
{schema}

Question:
{question}

Cypher:
{cypher}

Execution sample:
{rows}

Validation summary:
{validation}

Return strict JSON with:
- pass: boolean
- ambiguity_score: number from 0 to 1, lower is
better
- semantic_alignment_score: number from 0 to 1
```

```
- schema_use_score: number from 0 to 1
- difficulty: one of easy, medium, hard
- failure_reason: short string, empty if pass is
true
```

Pass only if the question is unambiguous, the Cypher answers it, the schema use is valid, and the result would be useful in an enterprise benchmark.

- Categorical property values in the schema constrain literal values written in the Cypher query.
- Do not reject because the execution sample returns a value that is absent from the categorical-value list; result rows are observed graph outputs.
- If deterministic validation says schema use is valid, lower schema\_use\_score only when the Cypher itself uses nonexistent schema elements, invalid relationship directions, or invalid literal values.

## H.2 Automation and Calibration

Table 32 gives the deployment contrast with Mind the Query, and Table 33 reports the completed human calibration packet. Human review has not disappeared from the research process. It has moved from a production gate to post-hoc calibration evidence.

| Dimension              | Mind the Query                    | PIPE-Cypher                               |
|------------------------|-----------------------------------|-------------------------------------------|
| Generation review gate | Manual logical review             | Deterministic gates + local LLM judge     |
| Human effort           | Reported 1,400 person-hours       | 80-row post-hoc judge calibration audit   |
| Private values         | Public benchmark values           | Configurable sampling and redacted export |
| Refresh                | Static dataset release            | Rerunnable private benchmark factory      |
| Model endpoint         | Gemini in their reported pipeline | Local Qwen3.5-9B endpoint                 |

Table 32: Industry deployment contrast with Mind the Query. PIPE-Cypher focuses on private refreshable benchmark generation rather than a one-time public dataset.

| Audit packet property      | Value               |
|----------------------------|---------------------|
| Rows                       | 80                  |
| Judge accept / reject      | 40 / 40             |
| FinBench / SNB rows        | 48 / 32             |
| Easy / medium rows         | 26 / 54             |
| Labeled rows               | 80                  |
| Calibration status         | complete            |
| Agreement / $\kappa$       | 0.800 / 0.600       |
| Judge precision (95% CI)   | 1.000 (0.912–1.000) |
| Judge recall (95% CI)      | 0.714 (0.585–0.816) |
| False-accept rate (95% CI) | 0.000 (0.000–0.138) |
| False-reject rate (95% CI) | 0.286 (0.184–0.415) |

Table 33: Post-hoc judge calibration packet coverage and agreement. Human labels calibrate the automated gate after generation and are not used as a generation gate.

## I Representative Accepted Examples

The examples below are selected in stable identifier order from the tracked full-export snapshot, one per graph/category cell when available. They show the natural-language question, accepted Cypher, structural tags, gate status, and a bounded execution-result sample.

1. **FinBench / Boolean Existence / easy. Question:** Does account '187743809466009406' have any outgoing transfer?

```
MATCH (src:Account {accountId:
'187743809466009406'})
-[:TRANSFER_TO]->
(:Account)
RETURN DISTINCT COUNT(DISTINCT src)
> 0
AS HasOutgoingTransfer
```

*Structure:* single\_hop, aggregation.

*Relationships:* TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {HasOutgoingTransfer: True}; observed rows: 1

2. **FinBench / Complex Aggregation / medium. Question:** What is the total transferred amount from accounts owned by person 'Gwar'?

```
MATCH (p:Person {personName:
'Gwar'})
-[:OWN_ACCOUNT]->
(src:Account)-[:TRANSFER_TO]->
(:Account)
RETURN DISTINCT SUM(t.amount) AS
TotalTransferredAmount
```

*Structure:* join\_heavy, aggregation.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {TotalTransferredAmount: 14972318.41}; observed rows: 1

3. **FinBench / Complex Retrieval / easy. Question:** Which accounts received transfers from accounts owned by person 'Zof'?

```
MATCH (p:Person {personName:
'Zof'})
-[:OWN_ACCOUNT]-> (src:Account)
-[:TRANSFER_TO]-> (dst:Account)
RETURN DISTINCT dst.accountId AS
AccountId, dst.accountType AS
AccountType, dst.isBlocked AS
IsBlocked
LIMIT 300
```

*Structure:* join\_heavy, bounded\_result.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4687402787162554854, AccountType: credit card, IsBlocked: False}; observed rows: 3

4. **FinBench / Negation Difference / medium. Question:** Which accounts owned by person 'Kant' have not sent any transfers?

```
MATCH (p:Person {personName:
'Kant'})
-[:OWN_ACCOUNT]-> (a:Account)
WHERE NOT (a) -[:TRANSFER_TO]->
(:Account)
RETURN DISTINCT a.accountId AS
AccountId, a.accountType AS
AccountType,
a.isBlocked AS IsBlocked
LIMIT 300
```

*Structure:* join\_heavy, negation, bounded\_result.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4739757132830738341, AccountType: merchant account, IsBlocked: False}; observed rows: 1

5. **FinBench / Path Temporal / medium. Question:** Which accounts can receive money within two transfer hops from accounts owned by person 'Sossamon'?

```
MATCH (p:Person {personName:
'Sossamon'}) -[:OWN_ACCOUNT]->
(src:Account) -[:TRANSFER_TO*1..2]->
(dst:Account)
RETURN DISTINCT dst.accountId AS
AccountId, dst.accountType AS
AccountType, dst.isBlocked AS
IsBlocked
LIMIT 300
```

*Structure:* join\_heavy, path, bounded\_result.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4687402787162554854, AccountType: credit card, IsBlocked: False}; observed rows: 4

6. **FinBench / Ranking Topk / medium. Question:** For accounts owned by person 'Barry', which account sent the highest total transfer amount?

```

1089 MATCH (p:Person {personName:
1090 'Barry'})
1091 -[:OWN_ACCOUNT]->
1092 (src:Account)-[:TRANSFER_TO]->
1093 (:Account)
1094 WITH src, SUM(t.amount) AS
1095 totalAmount
1096 RETURN DISTINCT src.accountId AS
1097 AccountId, src.accountType AS
1098 AccountType, src.isBlocked AS
1099 IsBlocked,
1100 totalAmount
1101 ORDER BY totalAmount DESC
1102 LIMIT 1

```

*Structure:* join\_heavy, aggregation, order\_rank, bounded\_result.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4732438783436260772, AccountType: internet account, IsBlocked: False}; observed rows: 1

#### 7. FinBench / Simple Aggregation / easy. Question: How many accounts are owned by person 'Kaewsuktae'?

```

1115 MATCH (p:Person {personName:
1116 'Kaewsuktae'}) -[:OWN_ACCOUNT]->
1117 (a:Account)
1118 RETURN DISTINCT COUNT(DISTINCT a) AS
1119 AccountCount

```

*Structure:* single\_hop, aggregation.

*Relationships:* OWN\_ACCOUNT.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountCount: 1}; observed rows: 1

#### 8. FinBench / Simple Retrieval / easy. Question: Which accounts are owned by person 'Barry'?

```

1128 MATCH (p:Person {personName:
1129 'Barry'})
1130 -[:OWN_ACCOUNT]-> (a:Account)
1131 RETURN DISTINCT a.accountId AS
1132 AccountId, a.accountType AS
1133 AccountType,
1134 a.isBlocked AS IsBlocked
1135 LIMIT 300

```

*Structure:* single\_hop, bounded\_result.

*Relationships:* OWN\_ACCOUNT.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4732438783436260772, AccountType: internet account, IsBlocked: False}; observed rows: 1

#### 9. SNB / Boolean Existence / medium. Question: Does person with id 6597069766828 like any post?

```

1146 MATCH (p:Person {id:
1147 6597069766828})
1148 OPTIONAL MATCH (p) -[:LIKES]->
1149 (post:Post)
1150 RETURN DISTINCT COUNT(post) > 0 AS
1151 LikesAnyPost

```

*Structure:* single\_hop, aggregation, optional.

*Relationships:* LIKES.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {LikesAnyPost: True}; observed rows: 1

#### 10. SNB / Complex Aggregation / medium. Question: How many distinct posts are in forums joined by person with id 6597069766845?

```

1161 MATCH (forum:Forum) -[:HAS_MEMBER]->
1162 (p:Person {id: 6597069766845})
1163 MATCH (forum) -[:CONTAINER_OF]->
1164 (post:Post)
1165 RETURN DISTINCT COUNT(DISTINCT post)
1166 AS
1167 JoinedForumPostCount

```

*Structure:* join\_heavy, aggregation.

*Relationships:* CONTAINER\_OF, HAS\_MEMBER.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {JoinedForumPostCount: 1}; observed rows: 1

#### 11. SNB / Complex Retrieval / easy. Question: Which people are members of forums containing posts tagged 'Manuel\_Noriega'?

```

1177 MATCH (forum:Forum) -[:HAS_MEMBER]->
1178 (p:Person), (forum)
1179 -[:CONTAINER_OF]->
1180 (post:Post) -[:HAS_TAG]-> (tag:Tag
1181 {name: 'Manuel_Noriega'})
1182 RETURN DISTINCT p.id AS PersonId
1183 LIMIT 200

```

*Structure:* join\_heavy, bounded\_result.

*Relationships:* CONTAINER\_OF, HAS\_MEMBER, HAS\_TAG.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {PersonId: 4398046511220}; observed rows: 19

#### 12. SNB / Negation Difference / easy. Question: Which person records are not linked from any message record through :HAS\_CREATOR?

|      |                                                              |
|------|--------------------------------------------------------------|
| 1193 | MATCH (p:Person)                                             |
| 1194 | WHERE NOT EXISTS((:Message)                                  |
| 1195 | -[:HAS_CREATOR]-> (p))                                       |
| 1196 | RETURN DISTINCT p.id AS PersonId,                            |
| 1197 | p.firstName AS PersonFirstName,                              |
| 1198 | p.lastName AS PersonLastName                                 |
| 1199 | <i>Structure:</i> single_hop, negation.                      |
| 1200 | <i>Relationships:</i> HAS_CREATOR.                           |
| 1201 | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                      |
| 1202 | <i>Result sample:</i> {PersonFirstName: R., Per-             |
| 1203 | sonId: 8796093022313, PersonLastName:                        |
| 1204 | Rao}; observed rows: 25                                      |
| 1205 | 13. <b>SNB / Path Temporal / easy.</b> <i>Question:</i>      |
| 1206 | Which people are within two knows hops of                    |
| 1207 | person with id 4398046511136?                                |
| 1208 | MATCH (src:Person {id:                                       |
| 1209 | 4398046511136})                                              |
| 1210 | -[:KNOWS*1..2]-> (dst:Person)                                |
| 1211 | RETURN DISTINCT dst.id AS PersonId,                          |
| 1212 | dst.firstName AS FirstName,                                  |
| 1213 | dst.lastName                                                 |
| 1214 | AS LastName                                                  |
| 1215 | LIMIT 200                                                    |
| 1216 | <i>Structure:</i> single_hop, path, bounded_result.          |
| 1217 | <i>Relationships:</i> KNOWS.                                 |
| 1218 | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                      |
| 1219 | <i>Result sample:</i> {FirstName: Rafael,                    |
| 1220 | LastName: Fernández, PersonId:                               |
| 1221 | 4398046511333}; observed rows: 25                            |
| 1222 | 14. <b>SNB / Ranking Topk / medium.</b> <i>Ques-</i>         |
| 1223 | <i>tion:</i> Which city records are linked                   |
| 1224 | from the most organisation records through                   |
| 1225 | :IS_LOCATED_IN?                                              |
| 1226 | MATCH (s:Organisation)                                       |
| 1227 | -[:IS_LOCATED_IN]-> (e:City)                                 |
| 1228 | WITH e, COUNT(DISTINCT s) AS                                 |
| 1229 | relatedCount                                                 |
| 1230 | RETURN DISTINCT e.id AS TargetId,                            |
| 1231 | e.name                                                       |
| 1232 | AS TargetName, relatedCount                                  |
| 1233 | ORDER BY relatedCount DESC                                   |
| 1234 | LIMIT 10                                                     |
| 1235 | <i>Structure:</i> single_hop, aggregation, or-               |
| 1236 | der_rank, bounded_result.                                    |
| 1237 | <i>Relationships:</i> IS_LOCATED_IN.                         |
| 1238 | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                      |
| 1239 | <i>Result sample:</i> {TargetId: 164, TargetName:            |
| 1240 | Kolkata, relatedCount: 130}; observed rows:                  |
| 1241 | 10                                                           |
| 1242 | 15. <b>SNB / Simple Aggregation / easy.</b> <i>Question:</i> |
| 1243 | How many posts are tagged 'Vietnam'?                         |
| 1244 | MATCH (post:Post) -[:HAS_TAG]->                              |
| 1245 | (tag:Tag                                                     |
| 1246 | {name: 'Vietnam'})                                           |
| 1247 | RETURN DISTINCT COUNT(DISTINCT post)                         |
| 1248 | AS                                                           |
| 1249 | PostCount                                                    |

|                                                        |      |
|--------------------------------------------------------|------|
| <i>Structure:</i> single_hop, aggregation.             | 1250 |
| <i>Relationships:</i> HAS_TAG.                         | 1251 |
| <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                | 1252 |
| <i>Result sample:</i> {PostCount: 1}; observed         | 1253 |
| rows: 1                                                | 1254 |
| 16. <b>SNB / Simple Retrieval / easy.</b> <i>Ques-</i> | 1255 |
| <i>tion:</i> Which post IDs did person with id         | 1256 |
| 4398046511124 like?                                    | 1257 |
| MATCH (p:Person {id:                                   | 1258 |
| 4398046511124})                                        | 1259 |
| -[:LIKES]-> (post:Post)                                | 1260 |
| RETURN DISTINCT post.id AS PostId                      | 1261 |
| LIMIT 200                                              | 1262 |
| <i>Structure:</i> single_hop, bounded_result.          | 1263 |
| <i>Relationships:</i> LIKES.                           | 1264 |
| <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                | 1265 |
| <i>Result sample:</i> {PostId: 343597385744}; ob-      | 1266 |
| served rows: 5                                         | 1267 |